Rajendran Prathibha Devkar

Dallas, TX • (682) 406-5414 • prathibhadevkar@gmail.com • linkedin.com/in/prathibha-devkar-3096

Software Engineer with 4+ years building full-stack applications and ML systems across enterprise analytics and research environments. Specializes in production debugging, real-time data visualization, and full-stack development across React, React Native, Node.js, and PyTorch. Ships production software used by enterprise clients, backed by independent research into applied ML systems.

EDUCATION

Master of Science in Computer Science — *The University of Texas at Arlington, USA* *May 2026*

Bachelor of Technology in Computer Science — *Saintgits College of Engineering, India* *May 2019*

EXPERIENCE

Research Assistant, SEAR Lab — *The University of Texas at Arlington* *Apr 2026 - Present*

- Refactored a 6-service Docker stack (Mosquitto MQTT, InfluxDB, Grafana, a FastAPI Data Hub, Python collectors) into a single self-contained FastAPI application with an embedded dashboard, eliminating the Docker/Grafana/MQTT infrastructure dependency entirely.
- Integrated live telemetry from three hardware APIs — LG ThinQ (AC), Rheem EcoNet (water heater), Enphase (solar) — into a unified REST API (/api/data, /api/history, /api/status) computing a real-time Net Load metric (solar minus HVAC/HWH draw), via a background asyncio collector refreshing every 60 seconds.

Software Developer Intern, Board Americas — *Remote, Texas* *May 2025 - Mar 2026*

- Resolved 45+ medium-to-high severity production bugs over 11 months across an enterprise forecasting platform (Aurelia/TypeScript, C#), improving platform stability across UI, state management, and backend API layers.
- Diagnosed and fixed critical D3.js charting bugs, including fiscal-year x-axis display errors and incorrect actual-vs-forecast rendering, correcting chart data clients relied on for business decisions.
- Fixed a company-wide cache-invalidation bug that let forecast data drift out of sync across users, by changing invalidation scope from per-user to per-company; also resolved a null-date bug across all datepicker components that was triggering 500 errors in production.

Technology Analyst, Infosys Private Ltd. — *Trivandrum, India* *Oct 2019 - Nov 2022*

- Developed React Native UI and integrated backend APIs for a curbside-pickup feature added to the Tractor Supply Company mobile app, improving user satisfaction by 30% based on pilot-release feedback.
- Built RESTful APIs in Node.js for an e-learning platform supporting 200+ concurrent users across student, teacher, and admin roles, enabling real-time session scheduling, participation, and new-user registration.
- Designed PostgreSQL schemas and optimized SQL queries for the platform; built a JMeter load-testing pipeline simulating 1K-10K virtual users, which identified and resolved a calendar API bottleneck for a team-reported 40% improvement in data-retrieval speed.
- Deployed the platform to live servers via PuTTY/WinSCP and managed process reliability with PM2, ensuring stable delivery across environments.

SKILLS

Databases: SQL Server, PostgreSQL, MySQL

Frontend: React, React Native, Aurelia, D3.js, HTML5, CSS3

Languages: Python, TypeScript, JavaScript, C#

ML: PyTorch, TensorFlow, Keras, OpenCV, NumPy, Pandas, CNN

AI/LLM: Ollama (local LLM inference), Prompt Engineering, DistilBERT fine-tuning, LLM evaluation & confidence calibration

Backend / APIs: ASP.NET, Node.js, FastAPI, RESTful APIs

Cloud: AWS, GCP

DevOps / Tools: Azure DevOps, Git/GitHub, Jira, Docker

PROJECTS

Apex F1 Predictor — *Python, FastAPI, scikit-learn, WebSocket*

- Built Apex F1 Predictor, a full-stack ML app forecasting Formula 1 race outcomes from historical race data (2023+) with a RandomForestClassifier and a FastAPI backend (autonomous retraining after each race, WebSocket-based live telemetry); reached AUC of 0.97/0.91/0.82 for win/podium/points-finish probability, though exact finishing-position prediction is harder (~4-position MAE) — a limitation surfaced directly in the app rather than hidden.

Recurrent Depth Transformer (RDT) — *PyTorch, Python*

- Implemented a Recurrent Depth Transformer from scratch in PyTorch, replacing stacked transformer layers with a single recurrently-looped block; matched a parameter-matched GPT baseline's accuracy on multi-hop reasoning (1-5 hops) at 30-47% fewer parameters depending on configuration, then ran inference-time loop-scaling experiments (6/10/14 loops) testing generalization to unseen chain lengths.
- Built and evaluated a tool-calling agent (3 tools, 17 test tasks) from the fine-tuned RDT model; caught a misleading evaluation metric that reported 100% task completion regardless of correctness, manually verified only 1-of-4 single-step answers were factually correct, and rewrote the evaluation harness to report results honestly instead of overstating them.

Confidence-Aware Meeting Intelligence — *Python, Ollama, DistilBERT, Streamlit*

- Built a confidence-aware meeting summarizer and action-item extractor that scores the reliability of each output instead of presenting everything with equal certainty, unlike Otter.ai/Fireflies-style tools (60-90% accuracy, no confidence signal): a local LLM (Ollama, llama3.2:3b) extracts summaries and action items with owner, due date, and a supporting quote, independently cross-checked by a separately fine-tuned DistilBERT classifier.
- Calibrated a confidence score (logistic regression on quote-grounding, owner-attribution plausibility, cross-model agreement, and self-reported confidence) on 228 hand-labeled examples; held-out accuracy: 58% for high-confidence items vs. 16% for low-confidence, a 3.6x spread confirming a genuine trust signal.
- Extracted the confidence-calibration logic into confidence-referee, a standalone MIT-licensed Python package (13 passing tests, CI across Python 3.9-3.12) that this project now depends on directly; validated it generalizes to a second, unrelated domain — RAG-answer hallucination detection on HaluEval — where its grounding signal alone reached 0.97 AUC and 94% held-out accuracy.

PUBLICATION

Devkar, R. Prathibha, Daniel, M.S., John, N.P., et al. (2021). Speaking Mouth System for Speech-Impaired People using Hand Gestures

Built a custom 4,000-image dataset (2,400 train / 1,600 test), architected a 2-layer CNN achieving 96% test accuracy, and implemented an end-to-end inference pipeline (gesture → text → synthesized speech); published in Springer FSST 2019.